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This is a reminder that both group and individual reports are due on March 2 at 12 noon. In your group report, please include a single page containing a brief statement on each group member's contribution to the report. A few sentences will be sufficient. Some examples:

1) AA worked on Sections xx, BB worked on Sections yy, CC worked on Sections zz. Everyone contributed to proof-reading and numerical work. 2) AA worked on method, BB worked on theory, CC worked on computation.

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Multiple Correction Method in FMRI - Draft

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-Date of handing in

February 9, 2026

1 Introduction

With the development of neuroimaging technology, Functional Magnetic Resonance Imaging (fMRI) has revolutionised our understanding of brain function by enabling the measurement of the activity of neurons through Blood Oxygen Level Dependent (BOLD) signals. However, typical fMRI analysis involves simultaneous hypothesis testing across $100 \times 100 \times 100$ voxels, creating a multiple comparison problem where false positive rates can reach unacceptably high levels. This would result in spurious brain activations that lead to incorrect scientific conclusions and skewed actual clinical applications.

In response, different multiple correction methods have been suggested to control false positives, yet these methods often face distinct limitations in their performance. For example, the Bonferroni correction, a traditional approach widely used in industry, is often criticized for being overly conservative, setting high thresholds that can miss true activations. Advanced methods such as Random Field Theory (RFT) and Permutation Tests have incorporated strong mathematical frameworks, but debates persist regarding their underlying assumptions and actual false positive control in practice. In advance, researchers have argued that these methods are rarely validated using real data, and have demonstrated that existing packages implementing RFT may produce false positive rates as high as 60-70% rather than the nominal 5%.

In this paper, we systematically examine Family-Wise Error Rate (FWER) control methods in fMRI analysis, focusing on both pixelwise and clusterwise inference using Random Field Theory and permutation testing approaches. Through simulations and applications to real datasets , we evaluate the practical performance of these methods under realistic conditions, assessing their effectiveness in controlling false positives.

Our goal is to provide practical guidance for researchers and clinicians on selecting appropriate multiple comparison correction strategies for their specific applications. In this paper, we present background information with literature review in Section 2 . Section 3 will outline our different methodology in controlling the false positive rates, and in sections 5 and 6 we will present our experimental results and discuss the implications of the proposed modifications.

2 Background

2.1 Contextual Background on FMRI

Functional Magnetic Resonance Imaging (fMRI) is one of the imaging methods used to investigate neuronal activity within the brain over time. Since its introduction in the 1990s, it has become a powerful tool in clinical research and cognitive neuroscience [1].

In fMRI, the Blood Oxygen Level Dependent (BOLD) signal is used as the target variable in measurement . This signal serves as a proxy for neuronal activity, allowing researchers to identify whether specific brain regions are stimulated by particular stimuli or tasks [2]. This is because when neurons in a brain region become active, there is an increase in cerebral blood flow that delivers oxygenated blood to the active area. This blood flow increase substantially exceeds the rise in oxygen consumption, resulting in a decrease in deoxyhemoglobin concentration [3, 4]. Since deoxyhemoglobin creates measurable distortions in the local magnetic field, its reduction produces an increase in the MRI signal, which can be detected and measured by fMRI scanners [5, 6].

In this project, we will focus on task-based fMRI, where participants in the scanner are required to perform specific tasks or are presented with stimuli that activate brain processes [7]. The experiment can be designed either as an event-related task where randomised stimuli are presented to participants at short and discrete time points, or as a block design where prolonged periods of stimulation alternate with rest or control conditions within each block [8]. During the experiment, brain images are captured repeatedly every few seconds throughout the session, which generate a time series of three dimensional brain volumes based on the BOLD signal.

After the fMRI experiment, the recorded data then undergo preprocessing stage to account for various inconsistency, sources and types of noise and artifacts [9]. These procedures includes motion correction to compensate for head movement [10], co-registration with spatial normalisation to align each participant's brain to a standard template (since individuals have different brain sizes and shapes) [11, 12], and prewhitening to remove non-white noise within the FMRI time series data [13].

Following data preprocessing, the analysis proceeds to statistical modeling. Here, a mass-univariate approach is adopted, in which the brain image is divided into very small three-dimensional units called voxels [14, 15]. Statistical tests are then performed independently at every single voxel in the brain image. Through implementation of the GLM and hypothesis testing, voxels showing BOLD responses that are significantly associated with the experimental conditions are identified (these statistical methods are discussed in detail in the following sections).

The mass-univariate approach, combining with statistical inferencing allows us to localise brain activation across the entire brain simultaneously. The voxels showing significant activation can then be marked and visualised in Neuroimaging Informatics Technology Initiative files (NIfTI) , which could store 3D maps commonly used for further analysis and interpretation in neuroscience research.

2.2 Statistical Modelling Framework

2.2.1 The General Linear Model (GLM)

To analyse the BOLD measurement at voxel level, the General Linear Model (GLM) is used as the basic statistical framework for analysing fMRI data [16]. Considering a dataset with n observations, the standard form of the GLM is expressed as:

$$Y = X\beta + \epsilon, \qquad \epsilon \sim N(0, \sigma^2 I), \tag{1}$$

where X is a $(n \times p)$ design matrix with p columns of predictors, and Y is the $(n \times 1)$ vector representing the measured BOLD activity across observations, β is the $(p \times 1)$ vector of regression coefficients with different catagories, and ϵ is the error vector that assumed to follow a normal distribution with variance $\sigma^2 I$.

The regression coefficients are estimated using ordinary least squares (OLS), yielding:

$$\hat{\beta}_{OLS} = (X^T X)^{-1} X^T Y, \tag{2}$$

with the variance estimate:

$$\hat{\sigma}^2_{OLS} = \frac{(Y - X\hat{\beta})^T (Y - X\hat{\beta})}{n}. \tag{3}$$

2.2.2 Time Series Assumption

Noting that fMRI data are time series in which the residuals typically exhibit temporal autocorrelation, such as AR or ARMA structure, we could not assume independence [13]. Therefore, we refined and modeled the error covariance structure as:

$$\mathrm{Var}(\epsilon) = \sigma^2 V. \tag{4}$$

Next, to apply OLS under these conditions, the model is transformed through a whitening procedure:

$$V^{-\frac{1}{2}} Y = V^{-\frac{1}{2}} X\beta + V^{-\frac{1}{2}} \epsilon, \qquad V^{-\frac{1}{2}} \epsilon \sim N(0, \sigma^2 I). \tag{5}$$

2.2.3 Statistical Inference and Hypothesis Testing

In order to identify brain pixels showing significant effects based on experimental conditions or subject characteristics, we conduct hypothesis testing by using contrast vectors L to specify particular effects of interest. For example, we could establish L to test whether covariates such as age or sex are associated with changes in the BOLD response when specific stimuli are activated.

The null and alternative hypotheses are formulated as:

$$H_0 : L\beta = 0 \quad \text{vs.} \quad H_1 : L\beta > 0 \tag{6}$$

The contrast vector L is a $(p \times 1)$ vector that defines specific linear combinations of the regression coefficients β. For example, to test whether the effect of condition A differs from condition B, we might specify $L = [1, -1, 0, \ldots]^T$, which tests the hypothesis $L\beta = \beta_1 - \beta_2$.

To evaluate these hypotheses, the t-statistic is computed at each voxel, which quantifies how many standard errors the estimated effect $L^T \hat{\beta}$ is away from zero under the null hypothesis.

$$t = \frac{L\hat{\beta}}{\sqrt{\mathrm{Var}(L\hat{\beta})}} \tag{7}$$

2.3 The Multi-Level Analysis Structure in FMRI

2.3.1 First-Level Analysis

First-level analysis, or subject-level analysis, is the stage of fMRI processing where the BOLD activity of each individual participant is modelled separately. For a given subject, BOLD images are collected across multiple runs, sessions, or time points, and the voxel-wise time series are analysed using the General Linear Model framework described in Section 2.2.

At the first level, the GLM is fitted independently at each voxel using the subject-specific design matrix X and observed BOLD time series Y. The model is given by

$$Y = X\beta + \epsilon, \tag{8}$$

Parameter estimation is performed using the whitened model described in Equation (5), yielding subject-specific estimates of the regression coefficients $\hat{\beta}$.

The estimated regression coefficients are assembled into voxel-wise beta maps, where each element of $\hat{\beta}$ quantifies the magnitude of the BOLD response associated with a particular experimental condition or covariate. Because brain structure and functional responses vary across individuals, fitting the GLM at the subject level is essential for capturing each participant's unique relationship between experimental events and observed BOLD signal changes.

To test specific hypotheses, contrast vectors L are applied to the estimated parameters, producing subject-level contrast images of the form $L\hat{\beta}$. Statistical inference is then conducted at each voxel using the corresponding t-statistics defined in Equation (7). These contrast images summarise the effect of interest for each subject and serve as the primary inputs to second-level analyses, where population-level inferences are made.

By ensuring that experimental effects are first estimated and tested within each individual, first-level analysis provides a statistically coherent foundation for subsequent group-level modelling and inference.

2.3.2 Second-Level Analysis

Second-level analysis, also referred to as group-level analysis, aims to make inferences about the population by combining first-level results obtained from numerous subjects. At the second level, statistical inference is performed by applying the general linear model independently at each voxel, across subjects rather than across time. Hence, while first-level analysis focuses on within-subject effects, second-level analysis addresses whether these effects are consistent across subjects and can therefore be generalised beyond the sampled individuals [17, 18].

At this stage, the input data consists of contrast images generated at the first level. Each contrast image represents a voxelwise estimate of a specific effect of interest for an individual. For a given voxel, these contrast values can be viewed as independent observations across subjects, and are entered into a group-level GLM, allowing systematic effects to be evaluated while accounting for between-subject variability. The resulting voxelwise statistical maps summarise the consistency of the estimated effects across subjects and form the basis for population-level inference. These group-level test statistic images are subsequently used for whole-brain inference, where appropriate correction for multiple comparisons is required. Unlike first-level fMRI time series data, second-level data does not exhibit autocorrelation across time, and it is therefore reasonable to assume that the error terms are independent and identically distributed (i.i.d.) [19, 20].

For a given voxel, the second-level model can be written as a General Linear Model of the form:

$$Y^{(2)} = X^{(2)}\beta^{(2)} + \varepsilon^{(2)}, \quad \varepsilon^{(2)} \sim \mathcal{N}(0, \sigma^2 I), \tag{9}$$

Where $Y^{(2)}$ is the $(n \times 1)$ vector of contrast values across n subjects at that voxel, $X^{(2)}$ is the second-level design matrix encoding group-level estimators, $\beta^{(2)}$ is the vector of group-level parameters, and $\varepsilon^{(2)}$ is the i.i.d. error term.

In the most straightforward case of a one-sample t-test, commonly used to assess whether an effect is consistently present across all subjects, $X^{(2)}$ reduces to a column of ones and $\beta^{(2)}$ represents the mean

effect at that voxel. Estimation is performed using ordinary least squares, and voxelwise t-statistics are computed in the same manner as in Equation (7), now reflecting between-subject variability rather than within-subject variability [21].

Although in this case the number of observations is reduced to the number of participants, the mass-univariate framework remains unchanged, and hypothesis testing is still carried out simultaneously across the entire three-dimensional brain volume. Testing thousands of voxels in parallel without correction would lead to an inflated number of false-positive activations; thus, second-level inference remains subject to a severe multiple comparisons problem [22, 23]. Consequently, controlling Family-Wise Error Rate at the group level becomes essential to ensure the validity of population-level results.

In this project, second-level analysis serves as the primary stage at which FWER correction methods are evaluated. Using voxelwise and clusterwise test statistics derived from group-level GLM, we apply Random Field Theory-based and permutation-based inference to control false-positive rates across the whole brain. By operating at the group level, these methods help to assess whether observed activations reflect genuine population effects rather than findings driven by noise or model assumptions.

2.4 The Multiple Comparisons Problem and Error Rate Control Methods

In the previous chapter, we demonstrated how the GLM is used to perform statistical tests in fMRI through the mass-univariate approach, which identifies and localises brain activation across the whole image. However, in practice the brain is divided into very small voxels, typically around 3 mm in width and with around $100 \times 100 \times 100$ in 3 dimensions [15], which introduces a significant statistical issue known as the Multiple Comparisons Problem (MCP). This section defines the MCP in the context of neuroimaging and outlines the theoretical frameworks used to control false-positive error rates.

2.4.1 Definition of the Multiple Comparisons Problem

In a standard fMRI analysis, the search volume usually has between $V = 50,000$ and $100,000$ voxels. This depends on the resolution and the brain mask used. For each voxel i, we test a null hypothesis H_{0i} (that there is no activation) against an alternative hypothesis. If we perform a single statistical test at a significance level of $\alpha = 0.05$, there is a 5% probability of rejecting the null hypothesis when it is true. This is known as a Type I error.

However, when we perform V independent tests at the same time, the probability of getting at least one false positive in the whole brain increases rapidly. This is called the Family-Wise Error (FWE). If we assume that the voxels are independent of each other, the probability of finding one or more false positives is:

$$P(\text{False Positives} \geq 1) = 1 - (1 - \alpha)^V. \tag{10}$$

For a typical brain where $V = 50,000$ and the uncorrected threshold is $\alpha = 0.05$, the probability becomes:

$$P(\text{False Positives} \geq 1) = 1 - (0.95)^{50,000} \approx 1.0. \tag{11}$$

This result shows that without correction, it is almost definite that we will find false brain activations, making the results scientifically invalid. For example, Bennett et al. illustrated this issue clearly by performing a standard fMRI analysis on a dead salmon. Because no multiple comparison correction was applied, they detected spurious significant activation in the salmon's brain cavity during the task[24]. This proved how easily noise can look like real signal in large datasets.

A major complexity in fMRI is that the data does not follow the independence assumption used in the equation above. The signals are spatially correlated. This happens for multiple reasons, such as: the working mechanism of the scanner, the nature of blood flow, and the Gaussian smoothing applied during preprocessing of images. This spatial correlation means that the real number of independent tests is lower than V, but the MCP remains a critical issue that still needs control.

2.4.2 Introduction of FWER and FDR

To solve the MCP, researchers must choose an error rate to control. The two main metrics used in FMRI are the Family-Wise Error Rate (FWER) and the False Discovery Rate (FDR).

Family-Wise Error Rate (FWER) The FWER is the probability of making one or more Type I errors in the entire set of tests. Control of the FWER needs:

$$\text{FWER} = P(N_{FP} \geq 1) \leq \alpha_{FWER}, \tag{12}$$

where N_{FP} is the number of false positives and α_{FWER} is the threshold (usually 0.05).

FWER is the standard method when we need high confidence in the specific location of activation.

False Discovery Rate (FDR) Unlike FWER, which tries to stop *any* false positives, FDR controls the proportion of false positives among the rejected hypotheses. If R is the number of active voxels and V_{FP} is the number of false positives, FDR is defined as:

$$\text{FDR} = E\left[\frac{V_{FP}}{R} \mid R > 0\right] \cdot P(R > 0). \tag{13}$$

The Benjamini- Hochberg (BH) procedure is the standard way to control FDR. This method is adaptive. If there is a lot of signal in the brain, FDR lowers the threshold to detect more voxels. This gives it higher sensitivity (power) than FWER. However, FDR is less specific about location. It guarantees that most of the active voxels are true, but it does not tell us exactly which ones are false.

Summary Clinical applications require high confidence in the exact location of detected activations. Therefore, this paper focuses on evaluating FWER control methods. We will compare the voxel-wise and cluster-wise performance of RFT against the performance of permutation testing.

3 Methodology

In this project, we will focus on multiple correction methods that control the FWER. We begin by introducing the Bonferroni correction, a simple and straightforward method that sets the significance threshold for each voxel to $\alpha/|V|$, such that:

$$\text{FWER} = \mathbb{P}(N_{FP} \geq 1) = \mathbb{P}\left[\max(p_v) \leq \frac{\alpha}{|V|}\right] = \alpha. \tag{14}$$

Although mathematically valid, this approach assumes that all tests are independent. However, because fMRI data exhibit strong spatial correlations, the Bonferroni correction is often overly conservative [25]. This resulted the significance threshold become stringent that true effects may be suppressed, leading to an increased rate of Type II errors in addition to reducing false positives [26].

Therefore, we introduce two more advanced methods: **Random Field Theory** and **Permutation Testing**, which are widely discussed in academia and industry and can improve precision while limiting the probability of obtaining at least one false positive. A brief explanation of these methods is provided below:

1. **Random Field Theory (RFT):** RFT addresses the problem of spatial correlation by using the Euler Characteristic (EC). Rather than treating the brain as a collection of independent voxels, it models the map of t-values as a continuous Gaussian random field. Dispite the fact that RFT does not correct for the number of voxels, it corrects for the number of resolution elements, or resels, in the image instead. In the context of FWER, it estimates the probability of observing clusters of a given size or height purely by chance.
2. **Permutation Testing:** Permutation methods are a non-parametric competitor of RFT. Instead of assuming a theoretical distribution, they build an empirical null distribution from the data itself. By randomly shuffling the labels of the design matrix (e.g. swapping "Patient" and "Control" labels), or applying sign flipping in case of a single regressor thousands of times, they remove the biological signal but keep the spatial correlation of the noise.

   For every shuffle, the maximum statistic is recorded (the maximum t-value or cluster size). The FWER threshold is then defined as the top 5% of this empirical distribution. Permutation testing is computationally expensive, but it is more reliable because it is not based on assumptions about normality or smoothness.

It is also important to note that the multiple comparison corrections mentioned above can be applied using either voxelwise methods or clusterwise methods. In this context, a FWER voxelwise approach aims to control the number of individual voxels that are false positives. In contrast, clusterwise methods take into account the fact that larger contiguous brain regions are more likely to contain true signals. Therefore, clusterwise correction groups together sets of adjacent voxels that exceed a specified threshold and evaluates the probability of such clusters being false positives.

The following sections, which describe the procedures, mathematical assumptions, and simulations for clusterwise and voxelwise methods under RFT and Permutation Testing, will be presented separately as an individual work. Please refer to the table below for our task distribution:

Name	Method
Lal	RFT (Voxelwise Method)
Joshua	RFT (Clusterwise Method)
Vlad	Permutation Test (Voxelwise Method)
Amari	Permutation Test (Clusterwise Method)

Table 1: FWER Methods discussed in subsequent sections

3.1 Voxelwise Permutation Testing

Random Field Theory controls false positives by relying on a strict set of assumptions about smoothness and Gaussianity. Permutation testing however constructs the null distribution directly from the observed data. It was first introduced to the neuroimaging public by Nichols and Holmes [21].

The main advantage of this method is that it does not rely on the assumption that the data is smooth enough or that the noise is strictly Gaussian. Instead, it relies on the assumption of *exchangeability* under the null hypothesis. In the context of FWER control, permutation testing provides exact control of the error rate, provided the exchangeability assumption holds.

3.1.1 Theoretical Framework: The Sign-Flipping Approach

To describe the permutation procedure formally, we use the standard General Linear Model framework introduced in Section 2.2. For a specific voxel v, let Y_v be the $N \times 1$ vector of observed contrast values for N subjects. The model is defined as:

$$Y_v = X\beta_v + \epsilon_v$$

where:

- X is the $N \times 1$ design matrix (a column of ones for a one-sample t-test).
- β_v is the parameter denoting the mean activation amplitude at voxel v.
- ϵ_v is the $N \times 1$ vector of residual errors.

We test the null hypothesis that there is no activation at voxel v ($H_0 : \beta_v = 0$). Under this hypothesis, the observed data consists only of the error component ($Y_v = \epsilon_v$).

The validity of the permutation test relies on the assumption of *exchangeability*. We assume that the distribution of error terms is symmetric about zero: $\epsilon \overset{d}{=} -\epsilon$. This implies that the probability of observing a value $+y$ equals the probability of observing $-y$. Thus, under H_0, the elements of the error vector are exchangeable with respect to their signs.

We can generate a valid realization of the data under the null hypothesis by element-wise multiplication of the data vector Y_v with a random sign vector $s \in \{-1, +1\}^N$:

$$Y_v^* = s \cdot Y_v$$

This sign-flipping procedure preserves the spatial correlation structure of the data within each subject's image (because the same scalar s_i is applied to all voxels of subject i simultaneously) and removes the systematic mean effect. This allows us to build a valid empirical null distribution.

3.1.2 Controlling FWER: The maxT Procedure

To solve the Multiple Comparisons Problem and control the FWER over the entire search volume $\mathcal{V}$, we cannot simply compute a threshold for each voxel independently. Instead, we need to control the probability of observing even one false positive anywhere in the brain:

$$FWER = \Pr(\max_{v \in \mathcal{V}} T(v) \geq u \mid H_0) \leq \alpha$$

To achieve this, we use the **maxT** procedure. For every permutation k, we calculate the t-statistic map over the whole brain and record only the maximum absolute t-value (u_k). This creates a null distribution of *maximum* statistics, which, by default, takes into account the spatial correlation structure of the data without needing the estimation of resolution elements as in RFT.

The algorithmic procedure is defined below:

Algorithm 1: maxT Permutation for FWER Control (One-Sample)

Input: Voxelwise data vectors $\{Y_v : v \in \mathcal{V}\}$, number of permutations K, significance level α
Output: Critical threshold t_{crit}

1. Compute the observed t-statistic map $T_{\text{obs}}(v)$ from the original data Y_v for all $v \in \mathcal{V}$.
2. **For** each permutation $k \in \{1, \ldots, K\}$:
   (a) Generate a random sign vector $s^{(k)} = (s_1, \ldots, s_N)^\top \in \{-1, +1\}^N$, where each s_i is drawn with probability 0.5.
   (b) Generate permuted data by sign-flipping:
   $$Y_v^{*(k)} = s^{(k)} \cdot Y_v \quad \text{for all } v \in \mathcal{V}.$$
   (c) Compute the t-statistic map $T_k(v)$ from $Y_v^{*(k)}$ for all $v \in \mathcal{V}$.
   (d) Record the maximum absolute statistic:
   $$u_k = \max_{v \in \mathcal{V}} |T_k(v)|.$$
3. Construct the empirical null distribution
   $$\mathcal{D}_{\text{null}} = \{u_1, u_2, \ldots, u_K\}.$$
4. Set the critical threshold t_{crit} as the $(1-\alpha)$-percentile of $\mathcal{D}_{\text{null}}$.
5. **Inference:** Mark voxel v significant if
   $$|T_{\text{obs}}(v)| > t_{\text{crit}}.$$

Exactness and Robustness A key advantage of this framework is that it provides exact (finite-sample) strong control of the FWER at level α, meaning the probability of a Type I error is no greater than α (up to the discreteness of the permutation distribution).

For example, Random Field Theory relies on asymptotic approximations involving the Euler Characteristic of the excursion set. These approximations are only accurate when the image is sufficiently smooth and the search volume is large. If the fMRI data has low smoothness or low degrees of freedom, RFT can become overly conservative or liberal. The permutation method changes its threshold automatically adjusting it to the actual smoothness of the noise field, making it robust to variations in preprocessing parameters such as the smoothing kernel size. This behaviour will be demonstrated in section 5.1.

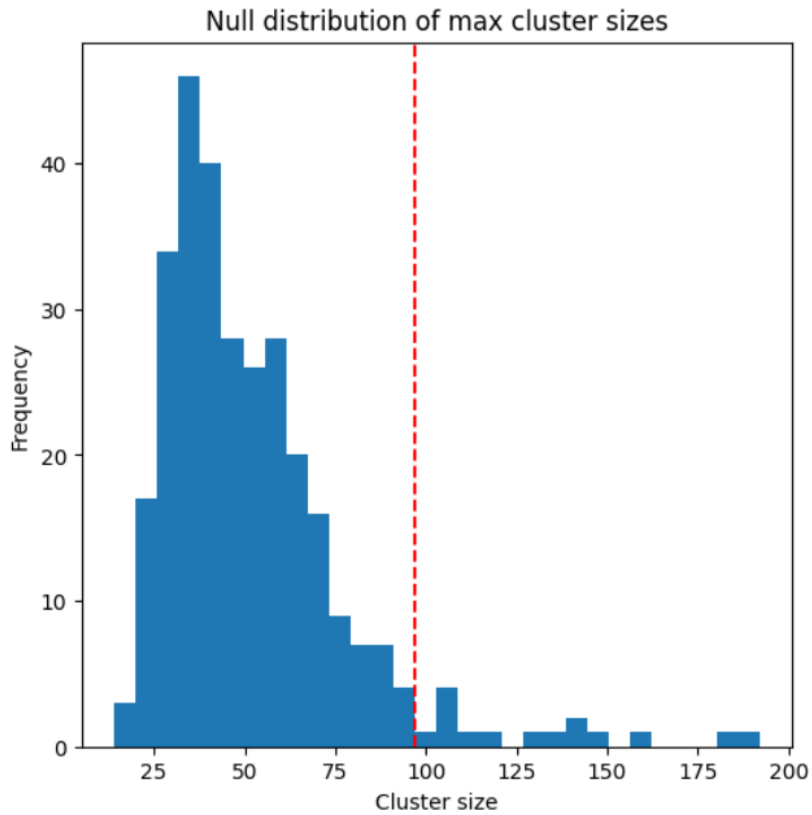
3.2 Clusterwise permutation test

Rather than testing individual voxels, suprathreshold cluster tests control the FWER in this case being the probability of observing at least one false positive cluster anywhere in the image. It does this by evaluating spatially contiguous groups of voxels that exceed a predefined statistical threshold. For each permutation, the largest suprathreshold cluster (the maximum connected set of voxels above the threshold) is recorded. Repeating this procedure across permutations yields a null distribution of the maximum cluster size under H_0. In the one-sample case, permutations are implemented by random sign-flipping of subject-level contrast images, which is valid under the null hypothesis of zero mean effect. This distribution is then used to assess whether the clusters observed in the original data are larger than would be expected by chance.

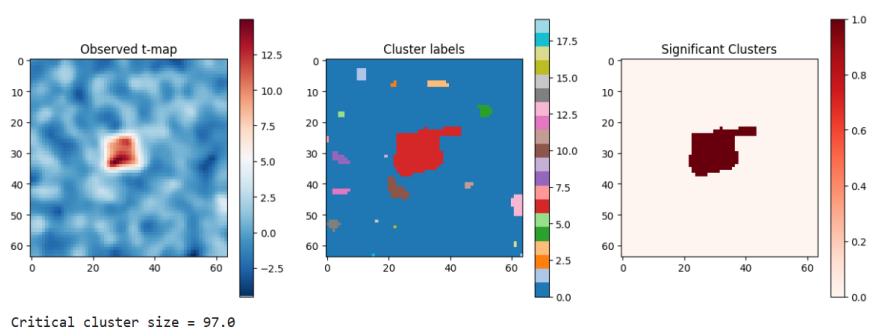
Algorithm 2: Clusterwise Permutation

1. Fit the voxelwise GLM
$$Y(v) = X\beta(v) + \varepsilon(v),$$
and estimate $\hat{\beta}(v)$ and the residual variance $\hat{\sigma}^2(v)$ using ordinary least squares.
2. Compute the observed t-statistic map for contrast L:
$$T_{\text{obs}}(v) = \frac{L^\top \hat{\beta}(v)}{\sqrt{\hat{\sigma}^2(v)\, L^\top (X^\top X)^{-1} L}}.$$
3. Form the observed suprathreshold mask:
$$M_{\text{obs}}(v) = \mathbb{I}\{T_{\text{obs}}(v) > u\}.$$
4. Label connected components of M_{obs} to obtain observed clusters $\{C_j^{\text{obs}}\}$ and their sizes $|C_j^{\text{obs}}|$.
5. **For** $k = 1$ to K:
   (a) Generate a random sign-flipping vector $s^{(k)} \in \{-1, +1\}^n$ and apply it to the data to obtain
   $$Y^{(k)}(v) = s^{(k)} \odot Y(v),$$
   where $\odot$ denotes element-wise multiplication.
   (b) Fit the GLM using $(Y^{(k)}, X)$ and compute the permuted t-statistic map $T_k(v)$ using the same contrast L.
   (c) Threshold the permuted map:
   $$M_k(v) = \mathbb{I}\{T_k(v) > u\},$$
   and label clusters $\{C_j^{(k)}\}$.
   (d) Record the maximum cluster size:
   $$m_k = \max_j |C_j^{(k)}|,$$
   with $m_k = 0$ if no suprathreshold clusters occur.
6. Sort $\{m_1, \ldots, m_K\}$ in ascending order: $m_{(1)} \leq \cdots \leq m_{(K)}$.
7. Define $c = \lfloor \alpha K \rfloor$ and set the critical cluster size:
$$c_{\text{crit}} = m_{(K-c)},$$

corresponding to the $(\lfloor \alpha K \rfloor + 1)$-th largest permutation maximum.



8. Declare an observed cluster C_j^{obs} significant if $|C_j^{\text{obs}}| \geq c_{\text{crit}}$. Define the significant cluster mask $M_{\text{sig}}(v) = 1$ for voxels belonging to significant clusters and 0 otherwise.



3.2.1 Assumptions

The key requirement for permutation validity is exchangeability under the null hypothesis. In the one-sample case, this is satisfied by sign-flipping: under the null hypothesis H_0 of zero mean effect, the distribution of subject-level contrast images is symmetric about zero, so randomly flipping their signs does not alter the joint distribution of the data. This assumption is typically reasonable where subject-level contrast images are treated as independent across subjects [21, 27]. The cluster-defining threshold u must be fixed, as the permutation procedure conditions on this choice.

FWER control comes from using the null distribution of the maximum cluster statistic. Under H_0, the

probability that any null realisation produces at least one suprathreshold cluster exceeding the critical cluster size c_{crit} is bounded by α by construction. Therefore the probability of observing one or more false positive clusters anywhere in the image is controlled at level α [21, 27].

4 Simulations

To evaluate the statistical properties of the voxelwise permutation method, we developed a Monte Carlo simulation engine. This allowed us to assess the method's FWER and Sensitivity under a certain set of conditions where the ground truth signal is known.

4.1 Data Generation and Signal Modeling

We simulated a set of $N = 20$ subjects to model a standard one-sample fMRI study. The data generation pipeline was created to reflect the physical properties and order of real fMRI data collection process, in particular the order in which neuronal activation and image formation and smoothing occur.

4.1.1 Noise and Spatial Correlation

For each subject i, a base noise field was generated by taking independent samples from a standard normal distribution $\mathcal{N}(0, 1)$ on a 64×64 grid. Here, ϵ_i denotes the unsmoothed noise image for subject i, and Y_i denotes the final simulated contrast image. To create realistic spatial covariance, we applied a Gaussian smoothing kernel G_σ:

$$\epsilon_i \sim \mathcal{N}(0, 1), \quad Y_{noise,i} = \epsilon_i * G_\sigma$$

We varied the smoothing parameter σ from 0.0 to 3.0 voxels. This step makes sure that our null data (H_0) accurately reflects the smoothness assumptions (or their lack of) tested by the method.

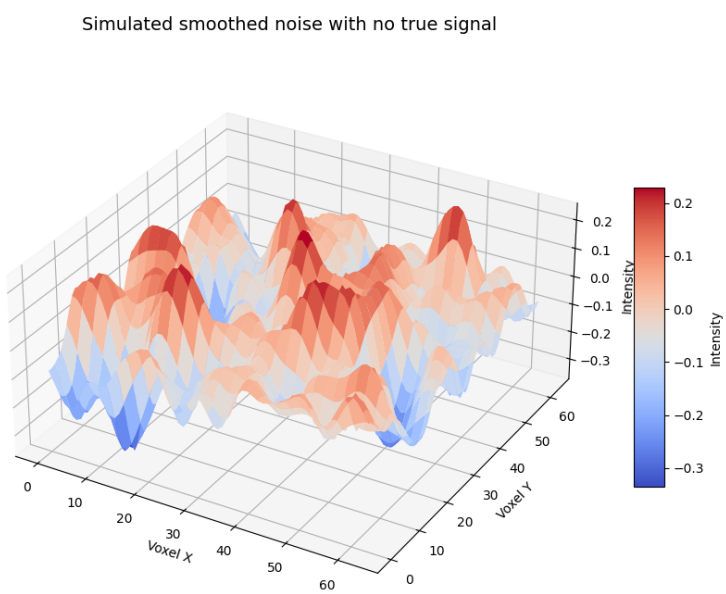


Figure 1: **Instance of the Noise Field.** A 3D surface plot of a realization of the noise field of a single subject after Gaussian smoothing applied. Although cluster- and peak- looking shapes are visible, they are purely random. To be valid, our correction method has to define a threshold high enough that these random peaks and valleys are not marked as significant activations more than $\alpha = 5\%$ of the time.

4.1.2 Signal Construction

An important feature of our simulation is that the activation source is defined before smoothing, and the resulting data contains a spatially smoothed version of this signal. We defined a circular neuronal activation source $\mathcal{M}$ with a fixed radius of 6 voxels in the centre of each subject's image. Under the alternative hypothesis (H_1), the raw data is modeled the following way:

$$Y_i = Y_{\text{noise},i} + Y_{\text{signal}}, \quad \text{where } Y_{\text{signal}} = (\lambda\, \mathbb{I}_{\mathcal{M}}) * G_\sigma.$$

where λ is the Signal-to-Noise Ratio (SNR). This creates realistic partial volume effects, where the signal intensity decays naturally at the boundaries. Because convolution with a Gaussian kernel is linear, the statement above is equivalent to smoothing the combined signal and noise field in one step:

$$(\epsilon_i + \lambda \mathbb{I}_{\mathcal{M}}) * G_\sigma = \epsilon_i * G_\sigma + (\lambda \mathbb{I}_{\mathcal{M}}) * G_\sigma.$$

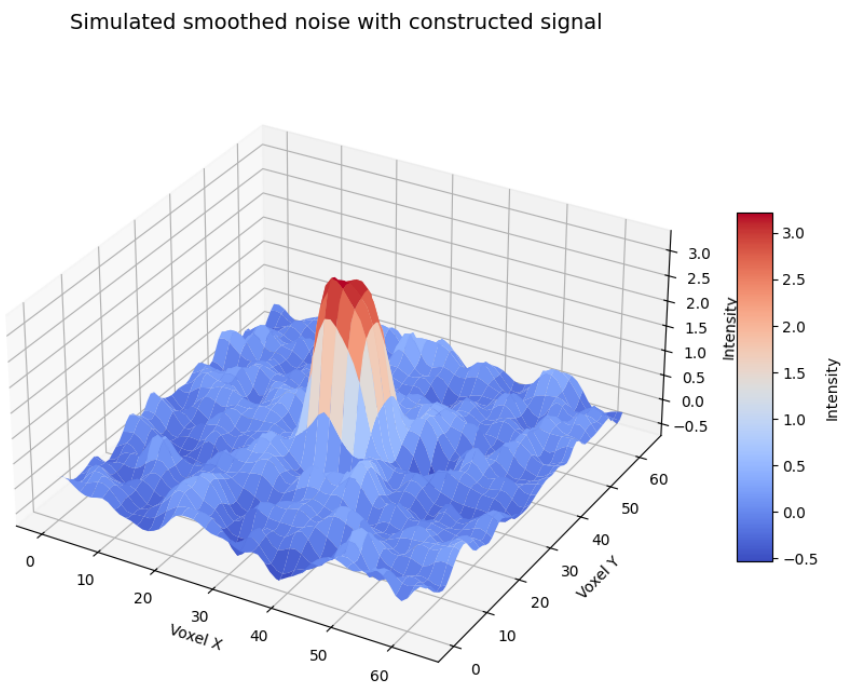


Figure 2: Example of a smoothed signal inserted in the center of a noisy randomly generated data, for smoothing sigma = 1.5 and SNR = 3

4.2 Defining the ground truth

A major challenge in verifying FWER on smoothed data is defining the exact boundary of the True signal. Because Gaussian smoothing spreads the signal energy beyond the original 6-voxel radius, strictly using the original binary mask would result in artificial false positives at the signal's edge (the signal bleeding effect).

To address this, we defined the **ground truth** by applying the simulation's smoothing kernel to the pure signal mask. As illustrated in Figure 3, the boundary for True Activation is defined where this smoothed signal profile remains non-zero.

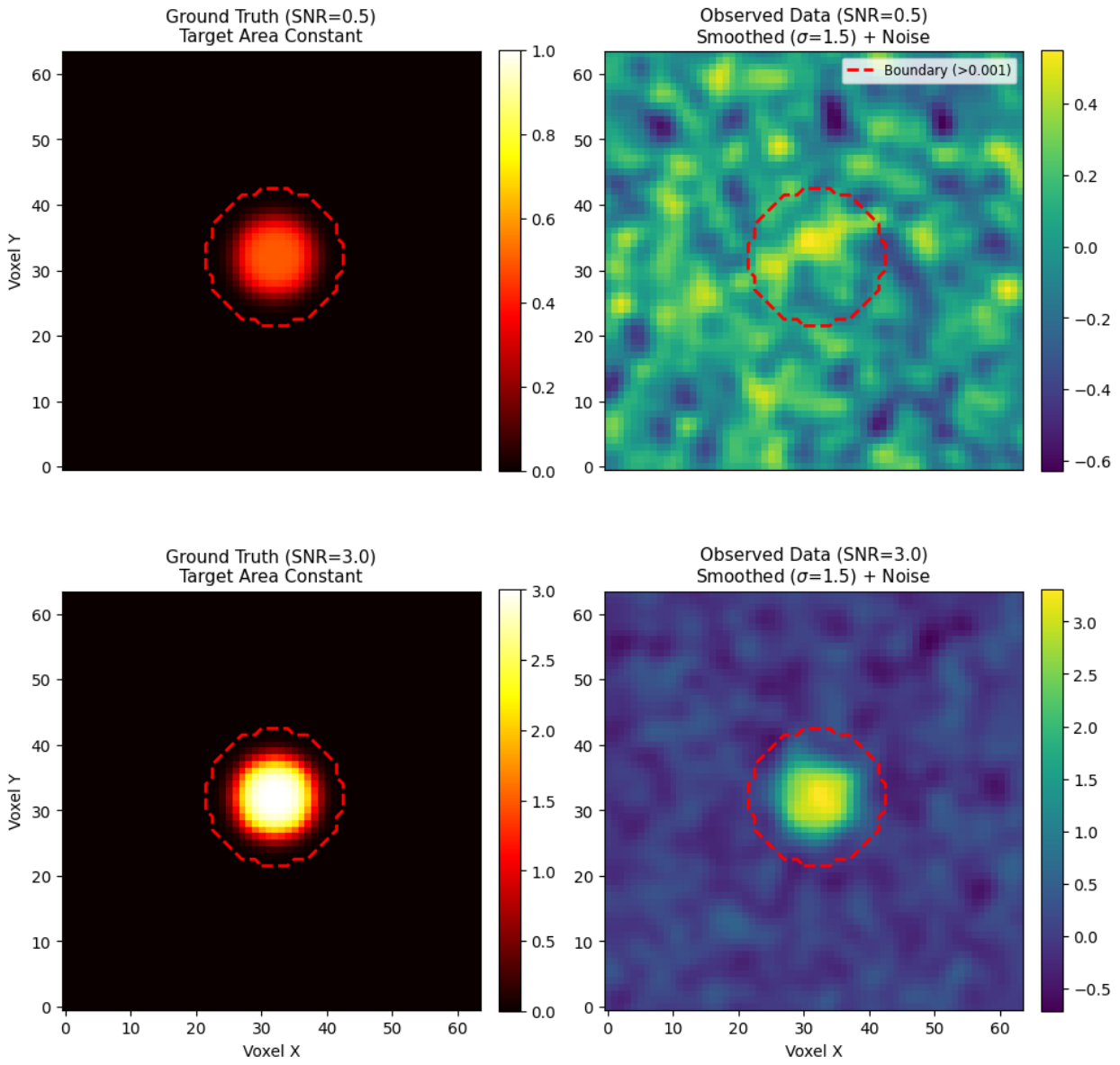


Figure 3: **Simulation Protocol and ground truth.** Left: The ground truth, defined by smoothing the discrete signal source. The red dotted line shows the boundary where signal exists. Right: A single realization of observed data (Noise + Signal + Smoothing). Note that the observed activation matches the ground truth boundary, assuring accurate FWER calculation.

Constructing the Evaluation Boundary To define the boundary shown in Figure 3 with a red dotted line, we ran a thresholding algorithm on the smoothed signal. Let S_{smooth} be the result of convolving the discrete 'cylinder' signal source with the smoothing kernel. The binary mask for the ground truth ($\mathcal{M}_{eff}$) is defined as the set of all voxels where the signal intensity is large enough:

$$\mathcal{M}_{eff} = \{v \in \mathcal{V} \mid S_{smooth}(v) > \epsilon\}$$

where ϵ is a small constant (set to 10^{-3} in our simulations). This approach ensures that the truth boundary dynamically adjusts with the smoothing σ, allowing us to accurately determine the full area of the signal's mask and prevent the misclassification of signal spread as background noise.

4.3 Performance Evaluation Metrics

Using the ground truth defined above, we calculated two key metrics:

- **FWER:** The probability of detecting significant activation **outside** the red boundary shown in Figure 3. If even a single voxel in the background noise is marked significant, the iteration is marked as a false positive event.
- **Sensitivity:** The proportion of voxels **inside** the boundary that are correctly identified as significant.

4.4 Parameter Sweep Strategy

To correctly evaluate the robustness of the permutation method compared to the other parametric approaches discussed in other sections of this paper, we performed a multi-dimensional parameter sweep:

1. **Smoothing (σ):** Varied from 0.0 to 3.0. This tests the method's ability to adapt to the data's potential smoothness without need for estimation of values such as Euler characteristics.
2. **SNR:** Varied from 0.0 to 3.0. This allows us to construct sensitivity curves and determine the detection threshold.

For each parameter combination, 200 Monte Carlo simulations were performed, with significance determined using the maxT permutation threshold derived from 500 permutations per run.

5 Results

5.1 Performance Evaluation: Voxelwise Permutation

We evaluated the performance of the voxelwise permutation method (Sign Flipping) using the One-Sample simulation framework described in Section 4. The results below summarize the method's ability to control false positives and detect true activation (Sensitivity) across a range of simulated conditions.

5.1.1 Control of FWER

The primary validity check for any multiple comparison correction is whether it maintains the false positive rate at or below the nominal significance level ($\alpha = 0.05$) **in the absence of signal**.

Figure 4 shows the empirical FWER calculated over 200 simulation runs with the full range of spatial smoothing levels ($\sigma \in [0.0, 3.0]$).

Robustness to Smoothness Assumptions A distinct advantage of the permutation approach observed here is its stability at low smoothing levels ($\sigma < 1.0$). Unlike Random Field Theory, which relies on strict approximations that often break down when the data is not smooth enough. In contrast, the permutation test builds the null distribution empirically from the data itself. Because of that, it successfully controls the error rate at 5% even when the data approaches white noise conditions ($FWER \approx 0.05$ at $\sigma = 0$). This confirms that the assumption of exchangeability is enough for this method to be valid regardless of spatial correlation.

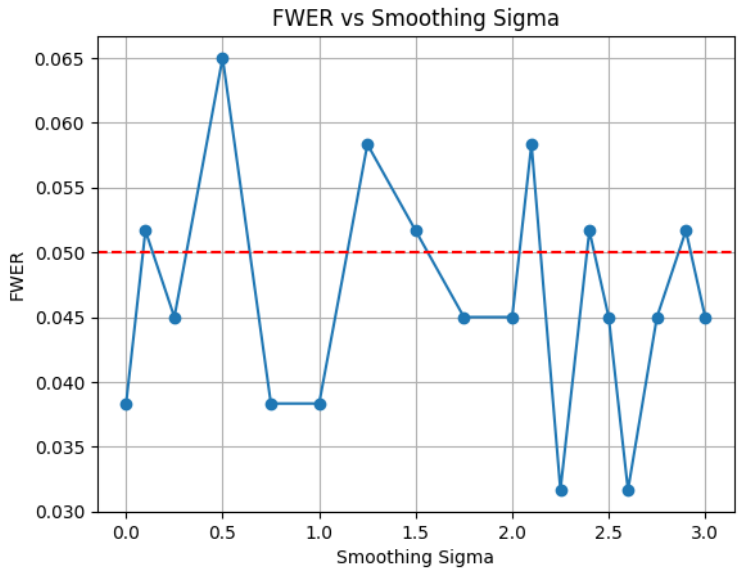


Figure 4: **FWER Control.** The empirical FWER plotted against spatial smoothing (σ). The red dashed line represents the nominal $\alpha = 0.05$ threshold. The method demonstrates robust control across all smoothness levels, oscillating stochastically around the target due to Monte Carlo variance.

5.1.2 Signal Analysis

After establishing that the voxelwise permutation method correctly controls the false positive rate in the absence of signal, we now construct a true activation signal and evaluate the method's performance across a range of SNRs and smoothing levels. Figure 5 provides a three-dimensional overview of these results. These 3D plots offer an intuitive global picture, however precise trends and comparisons are more clearly evaluated using two-dimensional slices, that are shown in the following subsections.

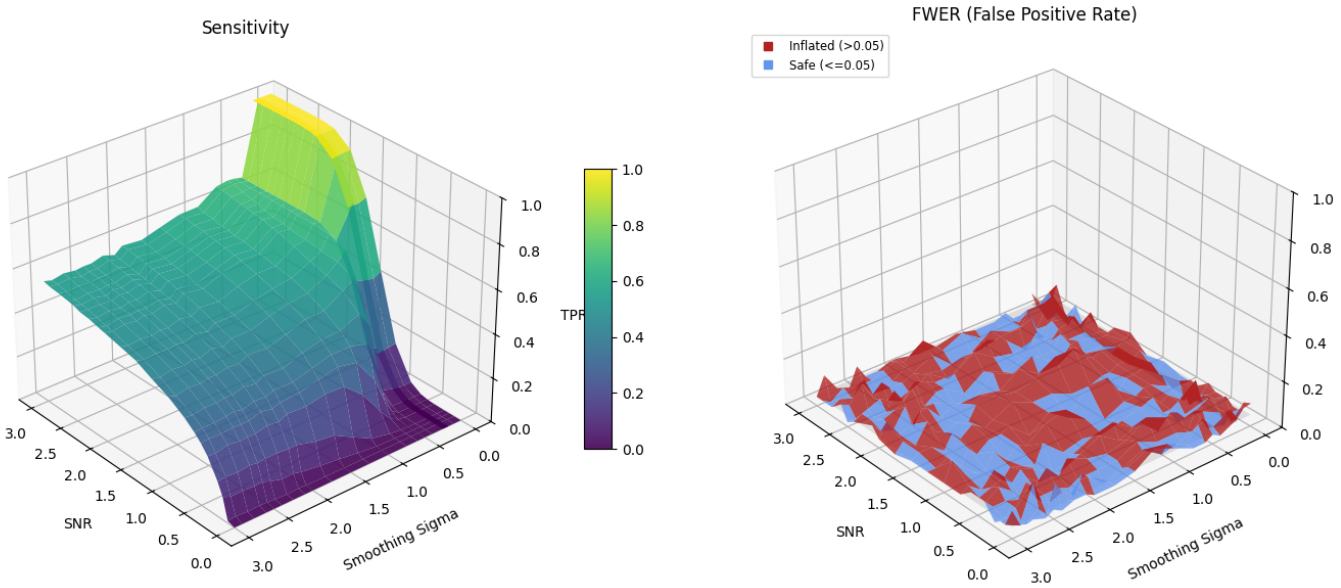


Figure 5: **Comprehensive Performance Landscape.**

5.1.3 Sensitivity Analysis under Signal Presence

We evaluated the method's sensitivity to detect the constructed signal over a range of varying SNRs and σ. Figure 6 displays the sensitivity curves for selected smoothing levels.

The results indicate a distinct sigmoidal response. We observe a strong interaction between smoothing and sensitivity:

1. **Effect of SNR:** Sensitivity exhibits a monotonic, sigmoidal dependence on SNR. As signal strength increases (moving right along the x-axis), the detection probability transitions from chance levels to 100%. For the voxelwise permutation method, the transition zone (where sensitivity rises most rapidly) is heavily dependent on the smoothness of the data.
2. **Effect of Spatial Smoothing:** Spatial smoothing significantly improves sensitivity when the SNR is low, has little effect at moderate SNR levels (SNR = 1.0), and can reduce sensitivity when the SNR is high. At low SNR, smoothing helps by reducing random noise, making weak signals easier to detect. This could be explained by the fact that at moderate SNR, the signal is already strong enough to be detected, so smoothing provides little additional benefit. At high SNR, excessive smoothing can diffuse the peak signal intensity into the surrounding background, potentially lowering the maximum t-statistic under the threshold.

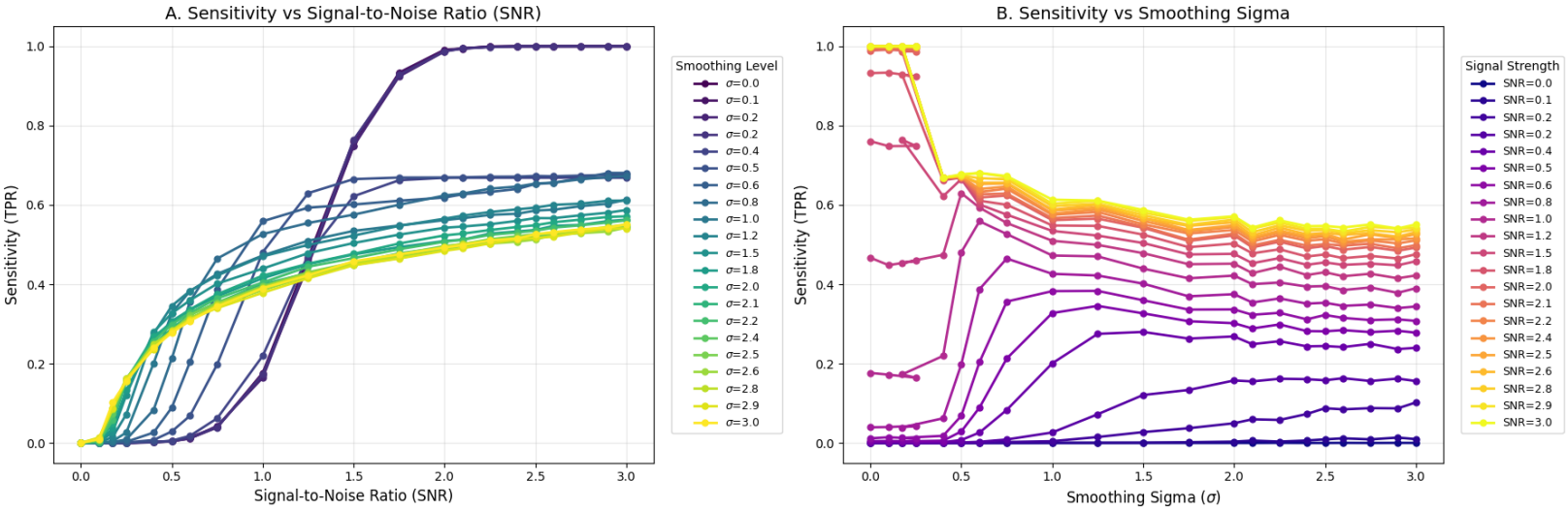


Figure 6: **Sensitivity Curves.** The True Positive Rate plotted against SNR. Different colored lines represent different smoothing levels. Higher smoothing (yellow/green lines) shifts the curve to the left, indicating higher sensitivity at lower SNR compared to raw data (dark purple lines), but also lower ceiling.

5.1.4 FWER Stability under Signal Presence

In addition to checking the method under the null hypothesis ($SNR = 0$), it is crucial to verify that the presence of a strong signal does not induce false positives in the surrounding non-active regions. This validates that the method correctly handles the ground truth and does not suffer from signal bleeding artifacts.

Figure 7 presents the FWER stability across varying conditions.

1. **Independence from Signal Strength (Panel A):** In theory, the false positive rate in the background noise should be independent of the signal strength in the active region. As shown in Panel A, the FWER fluctuates stochastically around 0.05 across the entire SNR range ($0.0 \rightarrow 3.0$). There is no

systematic upward trend. This confirms that defining the ground truth based on the smoothed signal profile correctly models the spatial extent of activation, preventing the misclassification of signal tails as false positives.

2. **Stability across Smoothing Regimes (Panel B):** Panel B shows that the presence of a strong signal ($SNR = 3.0$, red line) does not decrease the method's effectiveness. For both rough and smooth data, the permutation test successfully manages to adapt its threshold to separate the true activation in the ground truth mask (recall Section 4.2) from the background noise, maintaining the nominal 5% error rate.

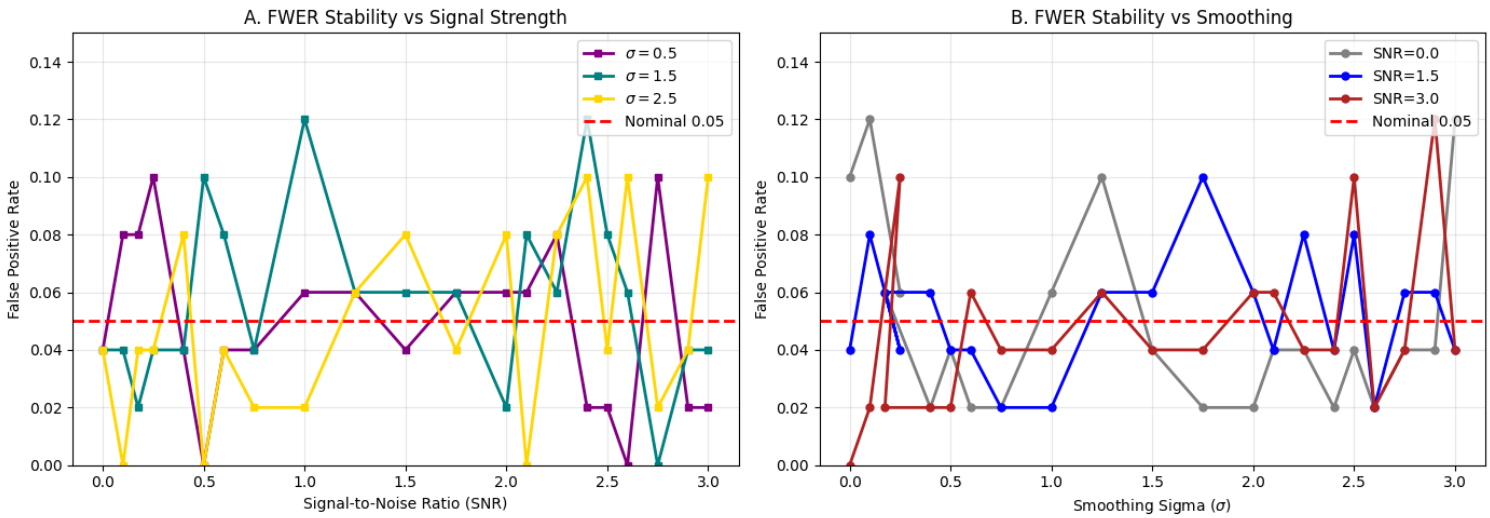


Figure 7: **Stability of Error Control. A:** FWER plotted against SNR for Low ($\sigma = 0.5$), Medium ($\sigma = 1.5$), and High ($\sigma = 2.5$) smoothing. The error rate remains close to the 0.05 line even as signal strength increases. **B:** FWER plotted against Smoothing for No Signal, Weak Signal, and Strong Signal. The method remains valid regardless of signal intensity.

5.1.5 Summary

The voxelwise permutation method proves to be a robust approach for the One-Sample design. It provides FWER control without requiring the data to meet Gaussian smoothness assumptions, making it a more reliable alternative to parametric methods in regimes of low spatial correlation, at a higher computational cost.

6 Discussion

7 Conclusion

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